

Double-Ended Queue (Deque) Introduction

Q. What is a Double-Ended Queue (Deque) Explain its types and how it differs from a standard queue and stack. Provide real-world applications.

[DEF] Definition to Remember

A Double-Ended Queue (Deque) is a generalized linear data structure where insertion and deletion of elements can be performed at both ends (Front and Rear). It effectively combines the capabilities of both a Stack (LIFO) and a Queue (FIFO) into a single versatile structure.

1. Operations in a Deque

Unlike standard queues (insert Rear, delete Front) or stacks (insert Top, delete Top), a Deque supports four primary operations:

1. Ins

Visual Representation:

Inse

2. Types of Restricted Deques

To enforce specific behaviors, Deques are classified into two restricted variants:

Type	Restriction	Allowed Operations
Input Restricted Deque	Insertion limited to ONE end	Insert (Rear only) Delete (Front &
Output Restricted Deque	Deletion limited to ONE end	Delete (Front only) Insert (Front &

3. Comparison with Stack and Queue

Data Structure	Insertion Point	Deletion Point	Principle
Stack	Top	Top	LIFO
Standard Queue	Rear	Front	FIFO
Deque	**Front & Rear**	**Front & Rear**	Both

4. Real-World Applications

* Web Browser History: When navigating back and forth, URLs are added/removed from one end. If history exceeds the max limit, oldest URLs are dropped from the other end.

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[MUST-WRITE] Must-Write Points (for 10 marks)

1. A Deque allows insertion and deletion at both ends (Front and Rear).
2. It combines the functionality of both Stacks and Queues.
3. Four main operations: Insert Front, Insert Rear, Delete Front, Delete Rear.
4. Input Restricted Deque: insertion at one end only, deletion at both.
5. Output Restricted Deque: deletion at one end only, insertion at both.
6. A standard queue restricts insertion to rear and deletion to front.

7. Key applications include Web Browser History, Undo/Redo, and the Work-Stealing scheduling algorithm.

[RECALL] Quick Recall

Deque -> Insert/Delete at BOTH ends -> Input Restricted (Insert 1 end) -> Output Restricted (Delete 1 end) -> Apps: Browser History, Undo/Redo, Work-Stealing